

46. CRC Calculator (CRCA)

The CRC (Cyclic Redundancy Check) calculator generates CRC codes.

46.1 Overview

Table 46.1 lists the specifications of the CRC calculator, and Figure 46.1 shows a block diagram of the CRC calculator.

Table 46.1 CRC Specifications

Item	Description
Data size	8 bits 32 bits
Data for CRC calculation*1	CRC codes are generated for any desired data in 8n-bit units (where n is a whole number) CRC codes are generated for any desired data in 32n-bit units (where n is a whole number)
CRC processor unit	8-bit parallel processing 32-bit parallel processing
CRC generating polynomial	One of three generating polynomials is selectable • 8-bit CRC $X^8 + X^2 + X + 1$ • 16-bit CRC $X^{16} + X^{15} + X^2 + 1$ $X^{16} + X^{12} + X^5 + 1$ One of two generating polynomials is selectable • 32-bit CRC $X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8$ $+ X^7 + X^5 + X^4 + X^2 + X + 1$ $X^{32} + X^{28} + X^{27} + X^{26} + X^{25} + X^{23} + X^{22} + X^{20} + X^{19}$ $+ X^{18} + X^{14} + X^{13} + X^{11} + X^{10} + X^9 + X^8 + X^6 + 1$
CRC calculation switching	The order of the bits produced by CRC calculation can be switched for LSB first or MSB first communication
Low power consumption	Module stop state can be set.

Note 1. The circuit does not have a function to divide data for calculation into CRC calculation units. Write data in 8-bit or 32-bit units.

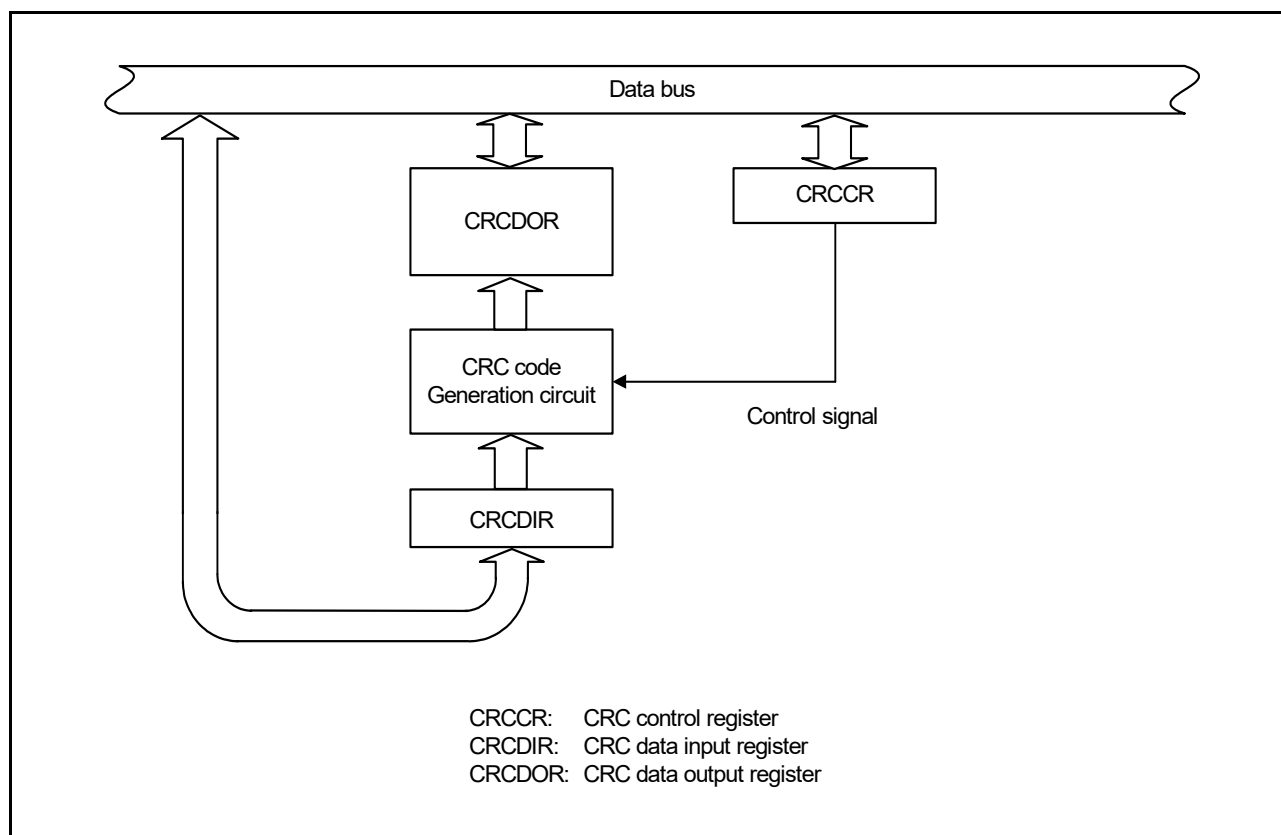


Figure 46.1 CRC Block Diagram

46.2 Register Descriptions

46.2.1 CRC Control Register (CRCCR)

Address(es): CRC.CRCCR 0008 8280h

b7	b6	b5	b4	b3	b2	b1	b0
DORCLR	LMS	—	—	—	GPS[2:0]		
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	GPS[2:0]	CRC Generating Polynomial Switching	b2 b0 0 0 0: No calculation is executed. 0 0 1: 8-bit CRC ($X^8 + X^2 + X + 1$) 0 1 0: 16-bit CRC ($X^{16} + X^{15} + X^2 + 1$) 0 1 1: 16-bit CRC ($X^{16} + X^{12} + X^5 + 1$) 1 0 0: 32-bit CRC ($X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1$) 1 0 1: 32-bit CRC ($X^{32} + X^{28} + X^{27} + X^{26} + X^{25} + X^{23} + X^{22} + X^{20} + X^{19} + X^{18} + X^{14} + X^{13} + X^{11} + X^{10} + X^9 + X^8 + X^6 + 1$) 1 1 0: No calculation is executed. 1 1 1: No calculation is executed.	R/W
b5 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b6	LMS	CRC Calculation Switching	0: Generates CRC for LSB first communication. 1: Generates CRC for MSB first communication.	R/W
b7	DORCLR	CRCDOR Register Clear	1: Clears the CRCDOR register. This bit is read as 0.	W*1

Note 1. Only 1 can be written.

LMS Bit (CRC Calculation Switching)

The setting this bit selects the order of the bits of generated CRC codes. The bit selects transmission of the lower-order byte of the CRC code first for LSB first communication, or the higher-order byte first for MSB first communication. For details on the transmission and reception of CRC codes, refer to **section 46.3, Operation**.

DORCLR Bit (CRCDOR Register Clear)

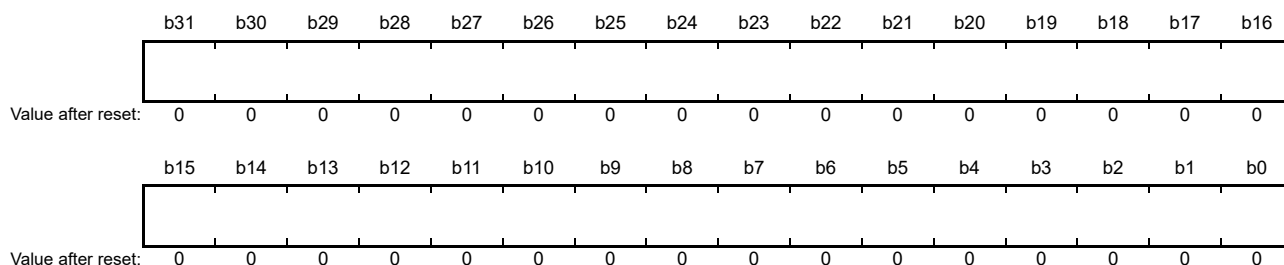
Write 1 to this bit so that the CRCDOR register is set to 0000 0000h.

This bit is read as 0. Writing 0 to this bit has no effect.

46.2.2 CRC Data Input Register (CRCDIR)

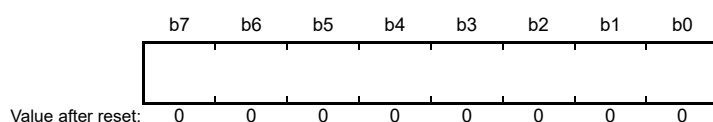
- When 32-bit CRC is selected

Address(es): CRC.CRCDIR 0008 8284h



- When 16-bit or 8-bit CRC is selected

Address(es): CRC.CRCDIR 0008 8284h



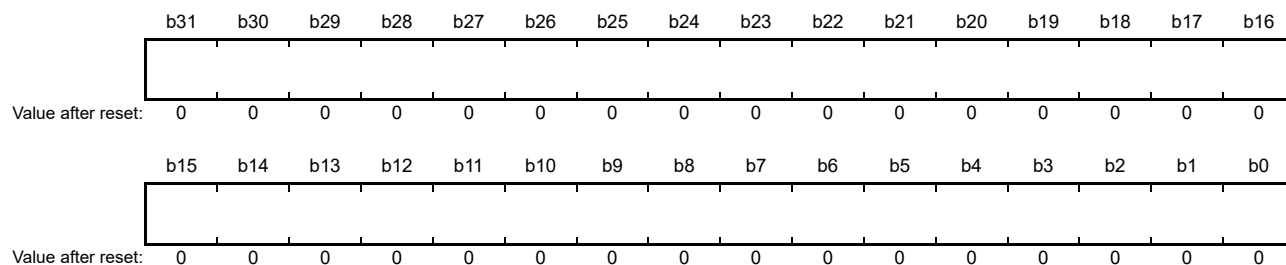
The CRCDIR register is a readable and writable register. Write data for CRC calculation to this register.

When generating a 32-bit CRC, the CRCDIR register should be accessed in longword units. When generating a 8-bit or 16-bit CRC, the CRCDIR register should be accessed in byte units.

46.2.3 CRC Data Output Register (CRCDOR)

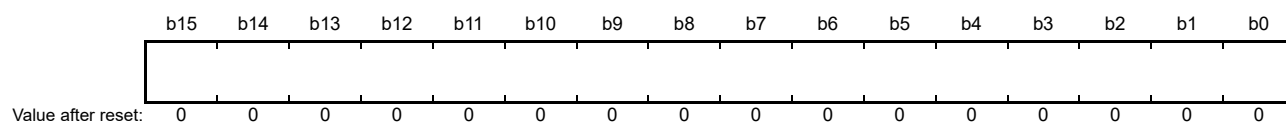
- When 32-bit CRC is selected

Address(es): CRC.CRCDOR 0008 8288h



- When 16-bit CRC is selected

Address(es): CRC.CRCDOR 0008 8288h



- When 8-bit CRC is selected

Address(es): CRC.CRCDOR 0008 8288h



The CRCDOR register is a readable and writable register.

The value after a reset is 0000 0000h. When calculating with the initial value set to a value other than 0000 0000h, rewrite the CRCDOR register.

Writing data to the CRCDIR register stores result of calculation in the CRCDOR register. In addition, following communication data, when a CRC code is written to the CRCDIR register and if the calculation result is 0000 0000h, there is no error in the communication data.

When a 32-bit CRC is selected, the CRCDOR register should be accessed in longword units. When a 16-bit CRC is selected, the CRCDOR register should be accessed in word units. When an 8-bit CRC is selected, the CRCDOR register should be accessed in byte units.

46.3 Operation

The CRC calculator generates CRC codes for use in LSB first or MSB first communication.

The following shows examples of generating the CRC code for input data (F0h) using the 16-bit CRC generating polynomial ($X^{16} + X^{12} + X^5 + 1$). In these examples, the value of the CRC data output register (CRCDOR) is cleared by setting the DORCLR bit to 1 to set the initial value for CRC calculation to 0000 0000h.

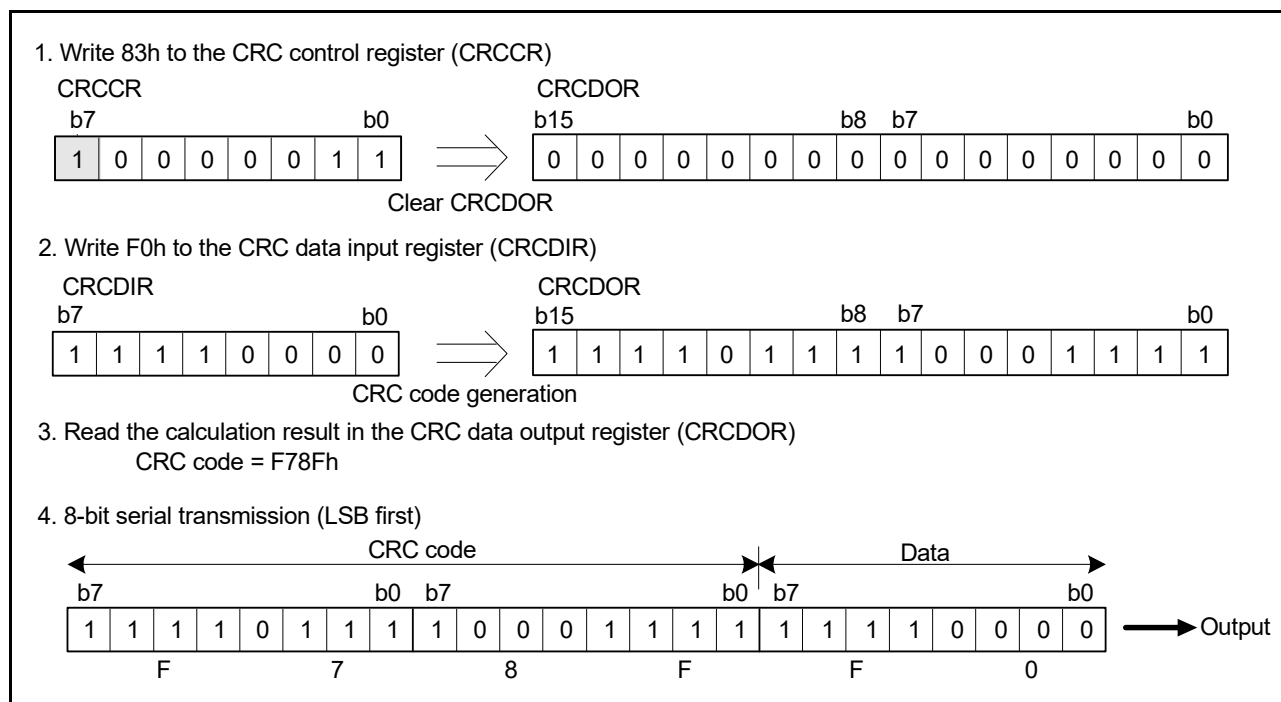


Figure 46.2 LSB First Data Transmission

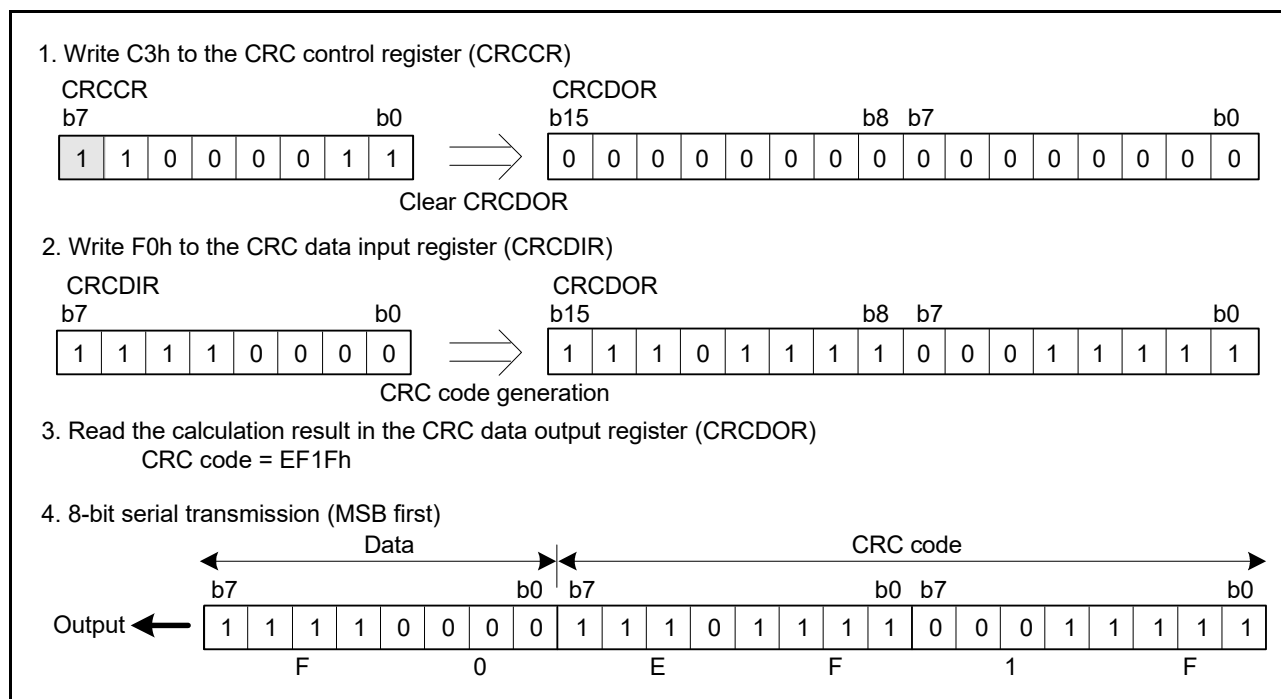


Figure 46.3 MSB First Data Transmission

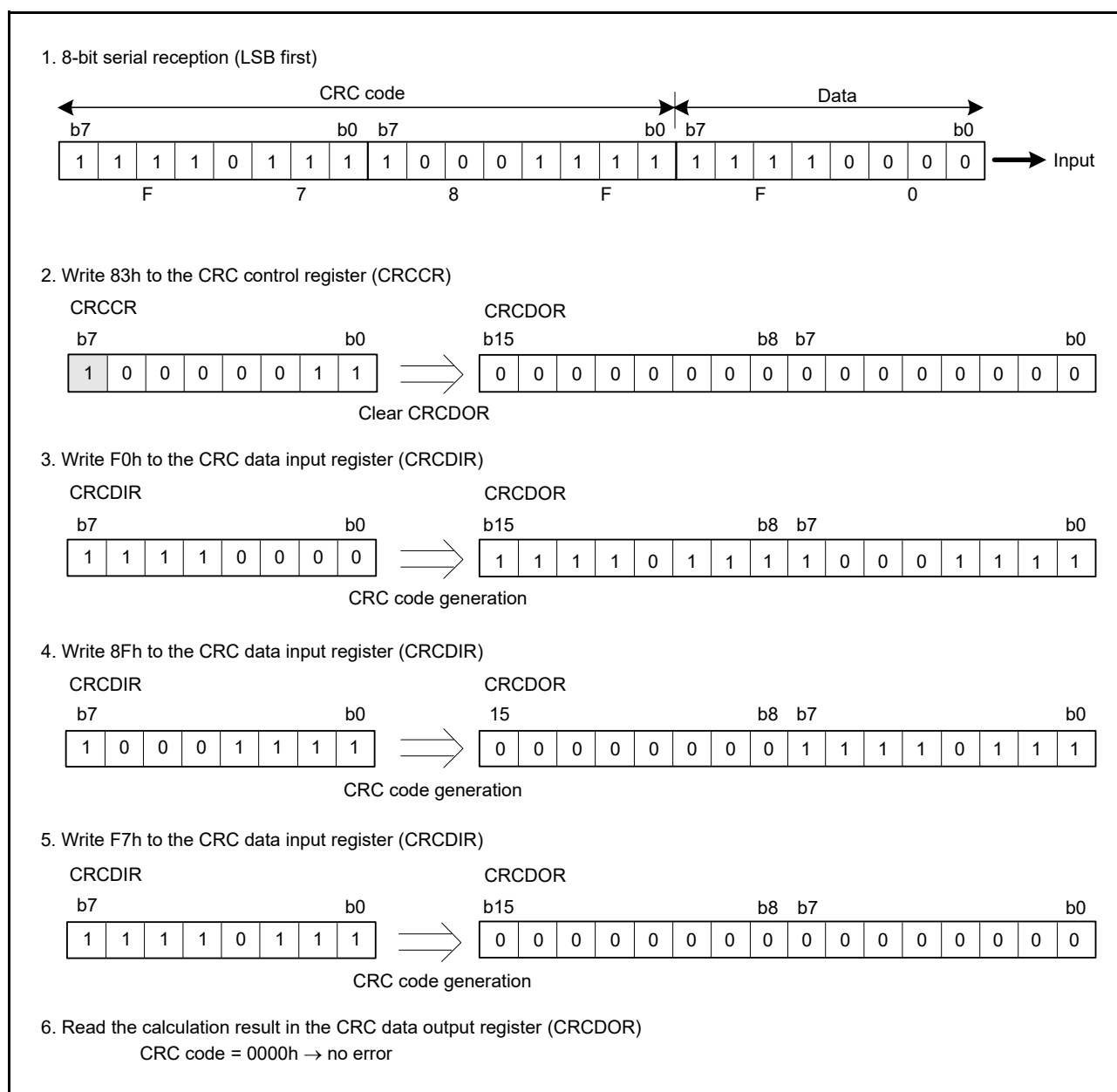


Figure 46.4 LSB First Data Reception

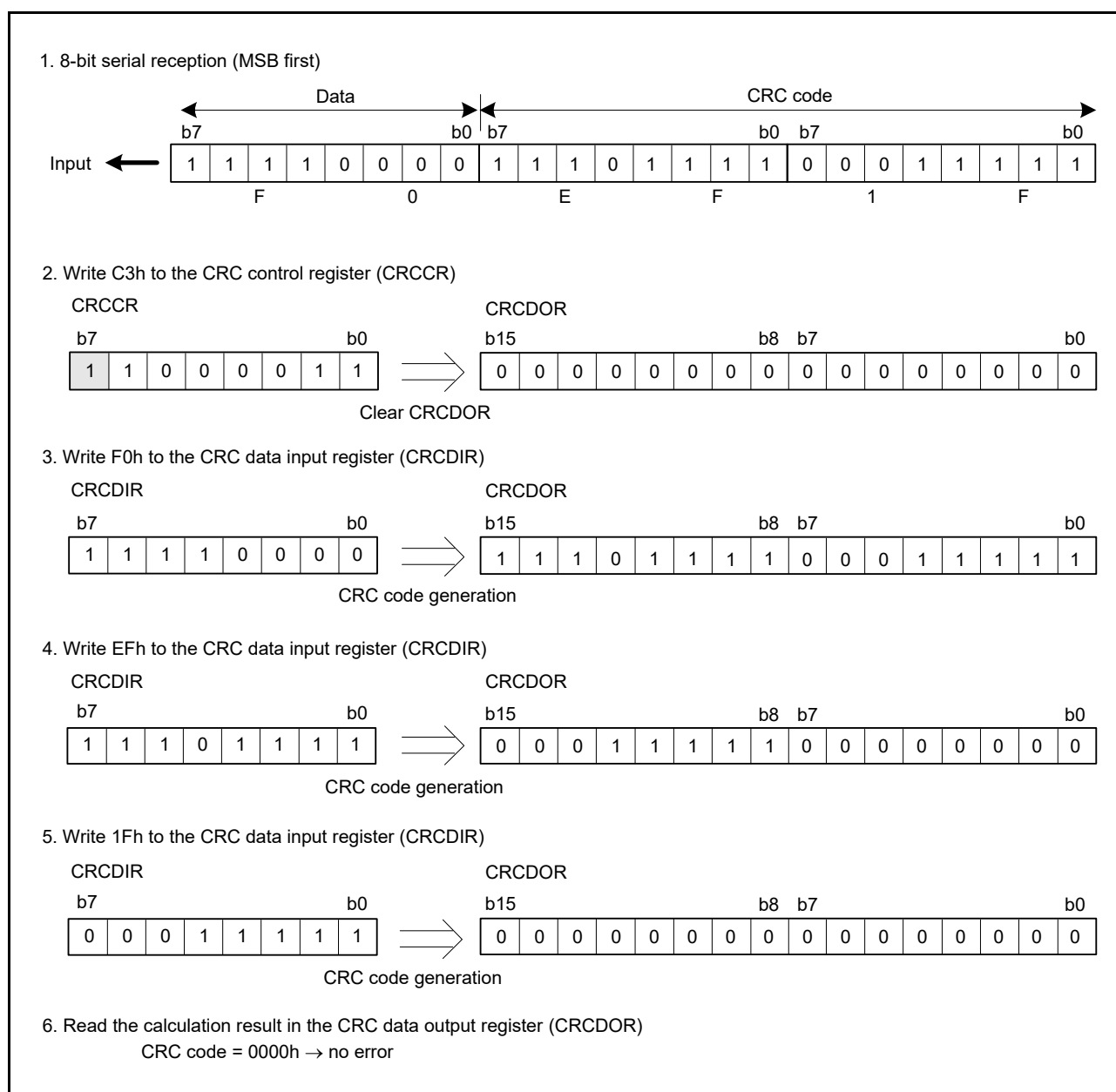


Figure 46.5 MSB First Data Reception

46.4 Usage Notes

46.4.1 Module Stop Function Setting

Operation of the CRC calculator can be disabled or enabled using the module stop control register B (MSTPCRB). After a reset, the CRC is in the module stop state. Register access is enabled by releasing the module stop state.

For details, refer to section 11, Low Power Consumption.

46.4.2 Note on Transmission

Note that the sequence of transmission for the CRC code differs according to whether transmission is LSB first or MSB first.

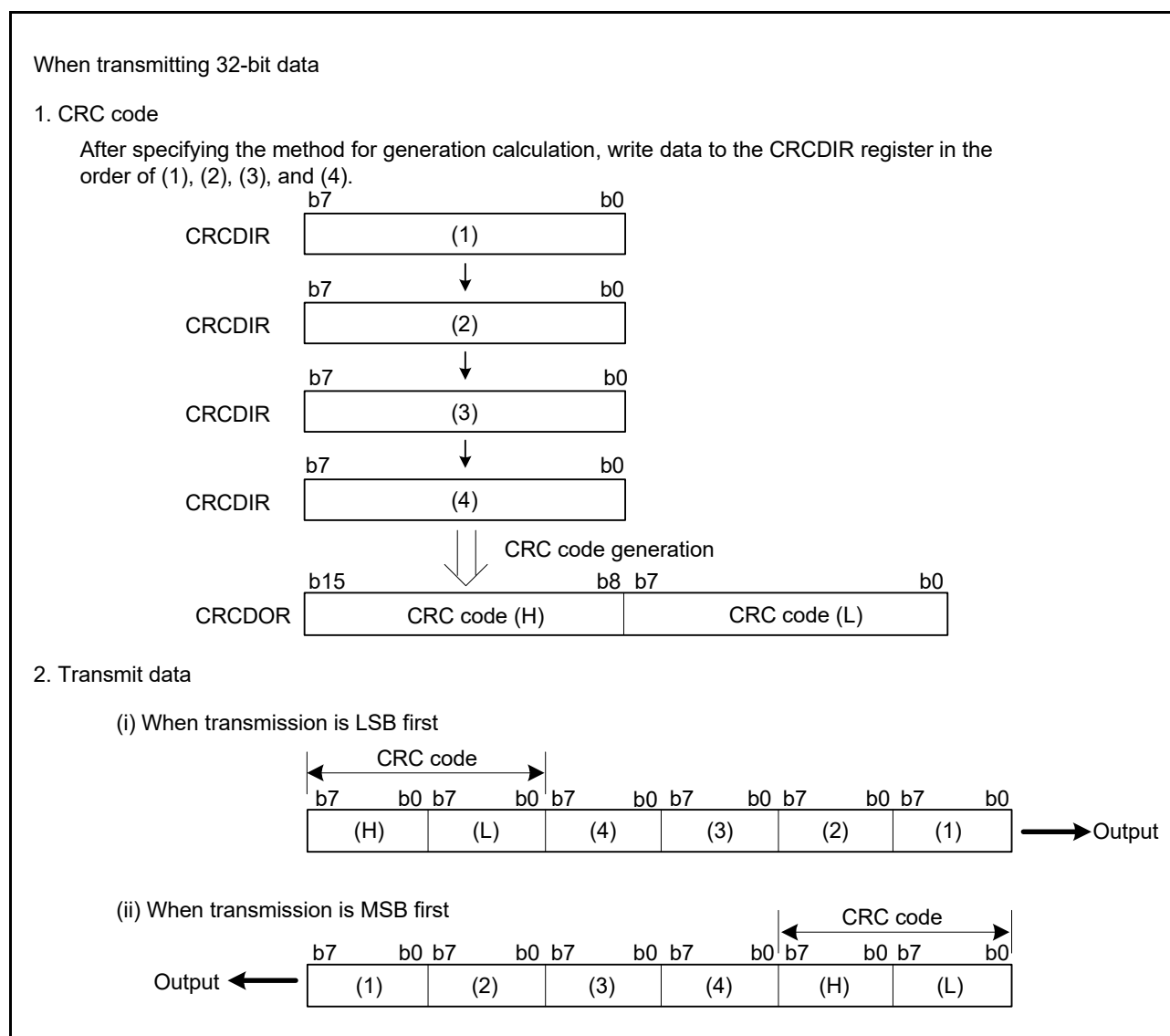


Figure 46.6 LSB First and MSB First Data Transmission